

LiveGS: Live Free-Viewpoint Video via High-Performance Gaussian Splatting for Mobile Devices

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Figure 1: LiveGS: Real-Time Data Acquisition, Rendering, and Mobile Display

Abstract

In this paper, we present LiveGS, an innovative real-time free-viewpoint video (FVV) live-broadcasting system that addresses key challenges in live volumetric video streaming. Our system demonstrates three significant technical advancements: (1) a novel view synthesis framework capable of generating high-fidelity volumetric human 3D Gaussian representations with real-time inference, eliminating the need for per-scene optimization and specifically designed for live streaming applications; (2) an efficient compression scheme that maps temporal 3D Gaussian representations onto 2D video planes, enabling transmission through standard video codecs with minimal quality degradation; and (3) a region-based Gaussian modeling strategy applies coarse granularity to low-frequency regions and fine granularity for high-frequency regions, significantly reducing computational complexity on mobile devices by

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SIGGRAPH Emerging Technologies '25, Vancouver, BC, Canada

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ACM ISBN 979-8-4007-1551-8/25/08

<https://doi.org/10.1145/3721257.3734033>

optimizing the number of 3D Gaussians required for accurate representation. The proposed system establishes a complete pipeline that enables real-time FVV playback on resource-constrained mobile devices while maintaining low transmission bandwidth requirements. Through extensive experimentation and system evaluation, we demonstrate that LiveGS represents the first practical solution for FVV live-broadcasting, featuring server-based temporal 3D Gaussian reconstruction, compression and transmission, while rendering can be performed on both PC and mobile platforms. This breakthrough significantly advances the state-of-the-art in mobile FVV systems by achieving an unprecedented balance between visual fidelity, computational efficiency, and transmission overhead.

ACM Reference Format:

Yuzhong Chen, Yuqin Liang, Zihao Wang, Danying Wang, Cong Xie, Shaohui Jiao, and Li Zhang. 2025. LiveGS: Live Free-Viewpoint Video via High-Performance Gaussian Splatting for Mobile Devices. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Emerging Technologies (SIGGRAPH Emerging Technologies '25)*, August 10–14, 2025, Vancouver, BC, Canada. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3721257.3734033>

1 Introduction

Real-time free-viewpoint video (FVV) enables immersive scene exploration from arbitrary angles, but achieving smooth playback on

mobile devices is challenging due to the high computational cost of volumetric reconstruction and bandwidth demands of high-fidelity 3D streaming. Existing methods often rely on offline processing, making them unsuitable for live mobile applications – for instance, [Jiao et al. 2024] relies on offline reconstruction and rendering of multiview perspective videos. In this paper, we present LiveGS, a novel live FVV system that supports real-time reconstruction, compression, and transmission. LiveGS enables seamless mobile playback, achieving over 30 FPS on an iPhone 15, with bitrates under 20 Mbps and latency under one second.

1.1 Real-time Reconstruction

3D Gaussian Splatting [Kerbl et al. 2023] has been widely used in novel view synthesis. However, it typically requires time-consuming per-scene optimization. To overcome this, we adopt a feedforward approach inspired by GPS-Gaussian [Zheng et al. 2024] to model the human body without per-scene tuning. We simplify further by using isotropic Gaussians, predicting only 1-scale and opacity for one Gaussian. We observe that flat, uniformly colored regions can be modeled with fewer Gaussians, and incorporate Sobel features with different granularity to adaptively model different body regions. This reduces Gaussian count by $\sim 70\%$, easing the rendering load on mobile devices (see Sec. 1.3).

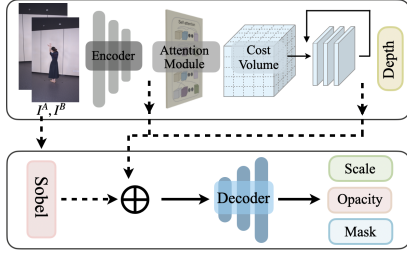


Figure 2: Pipeline of the proposed Gaussian Splatting method

The overview of our reconstruction method is illustrated in Fig. 2. Given stereo images, we extract features using a 2D CNN enhanced with attention modules [Ashish et al. 2017] to improve stereo matching by expanding the receptive field. A cost volume is built, and a GRU module [Lipson et al. 2021] iteratively refines the depth maps. We then fuse image, depth, and Sobel features using a U-Net-style decoder to regress Gaussian scale and opacity. Since the human body mostly exhibits diffuse reflection, we use the original RGB image as the color map for the Gaussians. Experiments confirm that our simplifications reduce the number of primitives significantly without significant degradation in rendering quality.

1.2 Low-loss Compression and Transmission

The pixel-aligned attributes of 3D Gaussians can be structured as conventional video streams, making them well-suited for compression using standard video codecs (see Fig. 3). To leverage this, we quantize and encode Gaussian attributes into the YUV 4:2:0 format. RGB values are transformed directly, while non-chromatic attributes, including scale, opacity, and mask, are linearly quantized to 8-bit and packed into the Y-channel. To preserve depth precision,

we apply Variable Bit Packing (VBP) [SIEKKINEN and KÄMÄRÄINEN 2023], and use morphological dilation on foreground depth maps to reduce compression artifacts near edges. We further opti-

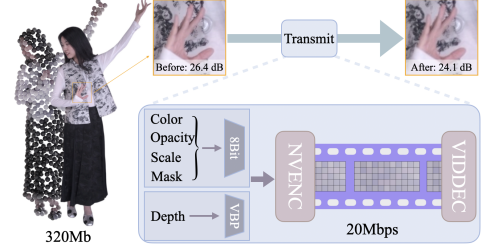


Figure 3: Illustration of encoding and transmission

mize compression by assigning bitrates based on each attribute’s impact on rendering quality. Depth and RGB, being most critical, receive higher bitrate via Region of Interest (ROI) encoding.

Our method achieves around $500\times$ compression, with rendered image PSNR loss kept below 3dB, striking a strong balance between visual quality and real-time transmission efficiency.

1.3 Efficient Mobile Rendering

The FVV player decodes the received 3D Gaussian attribute stream using a standard video codec, dequantizes the attributes, and reconstructs 3D Gaussians via 2D-to-3D projection using camera parameters. These Gaussians are then depth-sorted and rasterized in parallel, with novel views rendered through tile-based alpha blending. To reduce rendering load, the mask attribute guides Gaussian selection: all pixels in high-frequency regions are retained, while only every third pixel is retained in low-frequency areas. This reduces the number of active Gaussians by $\sim 70\%$, significantly improving efficiency on mobile devices.

Our mobile-optimized FVV player supports real-time interaction, including rotation, zoom, and translation, running at over 30 FPS on devices like the iPhone 15.

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